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The singular value decomposition

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Last time

- Two ways to compute $\mathbf{A} = \mathbf{QR}$: Gram-Schmidt, which loses orthogonality, and Householder reflections, which do not.
- How close the columns of $\mathbf{A}$ are to dependent: κ, the longest unit combination of the columns over the shortest.
- LU, Cholesky, QR each need something of $\mathbf{A}$: square, positive definite, full column rank.

Today: the decomposition that exists for every matrix, and gives κ its name. Then three things it is for: the closest solution, keeping only what matters, and finding the directions in data.

Part I · What the SVD is

Rotate, stretch, rotate

Every matrix has an SVD

Any real $m \times n$ matrix, of any shape and any rank, can be written

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

- $\mathbf{U}$ ($m \times m$) and $\mathbf{V}$ ($n \times n$) are orthogonal.
- $\mathbf{\Sigma}$ ($m \times n$) is diagonal, with the *singular values* $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.
- No conditions at all: unlike LU (square, pivoting), Cholesky (positive definite), or eigenvectors (which may not exist, or may not be orthogonal).

The columns of $\mathbf{U}$ and $\mathbf{V}$

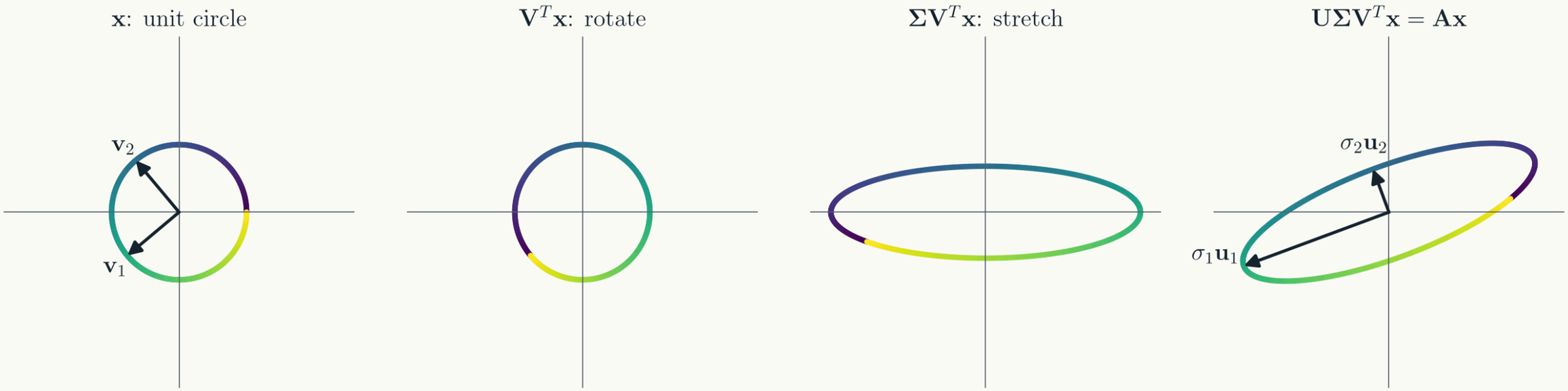
Name the columns: $\mathbf{V} = (\mathbf{v}_1\ \mathbf{v}_2 \cdots \mathbf{v}_n)$ and $\mathbf{U} = (\mathbf{u}_1\ \mathbf{u}_2 \cdots \mathbf{u}_m)$. Multiply $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ by $\mathbf{V}$ on the right, and $\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma}$. Column i of that reads

$$\mathbf{A}\mathbf{v}_i = \sigma_i\,\mathbf{u}_i$$

- The $\mathbf{v}_i$ are n perpendicular unit *input* directions, the $\mathbf{u}_i$ are m perpendicular unit *output* directions.
- $\mathbf{A}$ sends each $\mathbf{v}_i$ straight onto $\mathbf{u}_i$, stretched by σ_i. No mixing: $\mathbf{v}_1$ never leaks into $\mathbf{u}_2$.
- Any input is a combination of the $\mathbf{v}_i$, so this tells you what $\mathbf{A}$ does to everything.

That is the whole content of the SVD: the right pair of bases makes $\mathbf{A}$ a pure stretch.

What it means



- $\mathbf{V}^T$ rotates the columns $\mathbf{v}_i$ of $\mathbf{V}$ onto the axes; $\mathbf{\Sigma}$ stretches axis i by σ_i; $\mathbf{U}$ rotates the result to the output directions $\mathbf{u}_i$.
- The unit circle becomes an ellipse with semi-axes $\sigma_1\mathbf{u}_1$ and $\sigma_2\mathbf{u}_2$.

An example

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \end{pmatrix}$$

Its SVD is

$$\mathbf{U} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 0 & 0 & -\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \end{pmatrix}$$

The singular values are $4, 3, \sqrt{5}$ and 0. Each column of $\mathbf{V}$ is sent to σ_i times a column of $\mathbf{U}$: $\mathbf{A}\mathbf{v}_1 = 4\,\mathbf{u}_1$, and so on.

What the SVD reveals

- **Rank:** the number of nonzero singular values. The number of independent directions $\mathbf{A}$ can produce.
- **Range:** all the outputs $\mathbf{Ax}$ can reach. Spanned by the columns of $\mathbf{U}$ with $\sigma_i > 0$. A right-hand side $\mathbf{b}$ outside it has no solution.
- **Nullspace:** all inputs that $\mathbf{A}$ sends to zero. Spanned by the columns of $\mathbf{V}$ with $\sigma_i = 0$ (and any columns beyond m). Adding one to a solution gives another solution.
- **Conditioning:** the longest and shortest $|\mathbf{Ax}|$ over unit $\mathbf{x}$ are σ_1 and σ_n. Last lecture's κ is σ_1/σ_n: the *condition number*.

Rank plus nullspace dimension equals n, the number of unknowns.

Reading the example

- Three nonzero singular values, so the rank is 3.
- Range: columns 1 to 3 of $\mathbf{U}$. Any $\mathbf{b}$ with a component along column 4, $(0, 0, 1, 0)$, is out of reach.
- Nullspace: columns 4 and 5 of $\mathbf{V}$, $(-2, 0, 0, 0, 1)/\sqrt{5}$ and $(0, 0, 0, 1, 0)$. Any combination of them satisfies $\mathbf{A}\mathbf{x} = \mathbf{0}$.
- $\sigma_n = 0$, so $\kappa = \infty$: the columns of $\mathbf{A}$ are dependent.

This matrix gives all of it away by inspection: row 3 is zero, column 4 is zero, column 5 is twice column 1. A 1000×1000 matrix hides the same facts; the SVD digs them out.

SVD and eigenvalues

$\mathbf{A}$ need not be square, but $\mathbf{A}^T\mathbf{A}$ always is, and symmetric. Substitute the SVD:

$$\mathbf{A}^T\mathbf{A} = \mathbf{V}\boldsymbol{\Sigma}^T\mathbf{U}^T\,\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T = \mathbf{V}\left(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\right)\mathbf{V}^T$$

- That is an eigen-decomposition: the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}^T\mathbf{A}$, with eigenvalues σ_i^2. For our example, $16, 9, 5, 0, 0$.
- Likewise $\mathbf{A}\mathbf{A}^T = \mathbf{U}\left(\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\right)\mathbf{U}^T$: the columns of $\mathbf{U}$ are its eigenvectors.
- If $\mathbf{A}$ is itself symmetric, its singular values are $|\lambda_i|$, the absolute values of its eigenvalues.

This is why the SVD always exists: a symmetric matrix always has an orthonormal set of eigenvectors. It is not how the SVD is computed; that comes at the end.

Part II · The closest solution

When $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no solution

Solving with the SVD

If $\mathbf{A}$ is square with all $\sigma_i > 0$, the inverse is three easy moves, because orthogonal matrices are inverted by transposing and a diagonal one by reciprocals:

$$\mathbf{A}^{-1} = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)^{-1} = \mathbf{V}\,\mathbf{\Sigma}^{-1}\mathbf{U}^T, \qquad \mathbf{\Sigma}^{-1} = \operatorname{diag}(1/\sigma_i)$$

rotate $\mathbf{b}$ into the output directions	$\mathbf{c} = \mathbf{U}^T\mathbf{b}, \quad c_i = \mathbf{u}_i \cdot \mathbf{b}$
undo each stretch	$y_i = c_i/\sigma_i$
rotate back to the input directions	$\mathbf{x} = \mathbf{V}\mathbf{y} = \sum_i y_i\,\mathbf{v}_i$

Each move is exact up to rounding. Nothing is divided by a pivot that elimination happened to produce; the only divisions are by the σ_i, which are all known before the first division is done.

When some σ_i are zero

- Step two divides c_i by σ_i. If $\sigma_i = 0$, the direction $\mathbf{u}_i$ is outside the range: no $\mathbf{x}$ can produce that part of $\mathbf{b}$. The best we can do is leave it, and set $y_i = 0$.
- That defines the *pseudoinverse*:

$$\mathbf{A}^{+} = \mathbf{V}\,\mathbf{\Sigma}^{+}\mathbf{U}^{T}, \qquad \mathbf{\Sigma}^{+} = \mathrm{diag}\big(1/\sigma_i \text{ if } \sigma_i > 0, \text{ else } 0\big)$$

  It exists for every matrix, square or not.

- $\mathbf{x} = \mathbf{A}^{+}\mathbf{b}$ **minimizes** $|\mathbf{A}\mathbf{x} - \mathbf{b}|$. If several $\mathbf{x}$ tie (a nullspace), it is the shortest of them.

In practice "zero" means "below a cutoff": a σ_i of $10^{-15}\sigma_1$ is rounding noise, and dividing by it would amplify noise by 10^{15}.

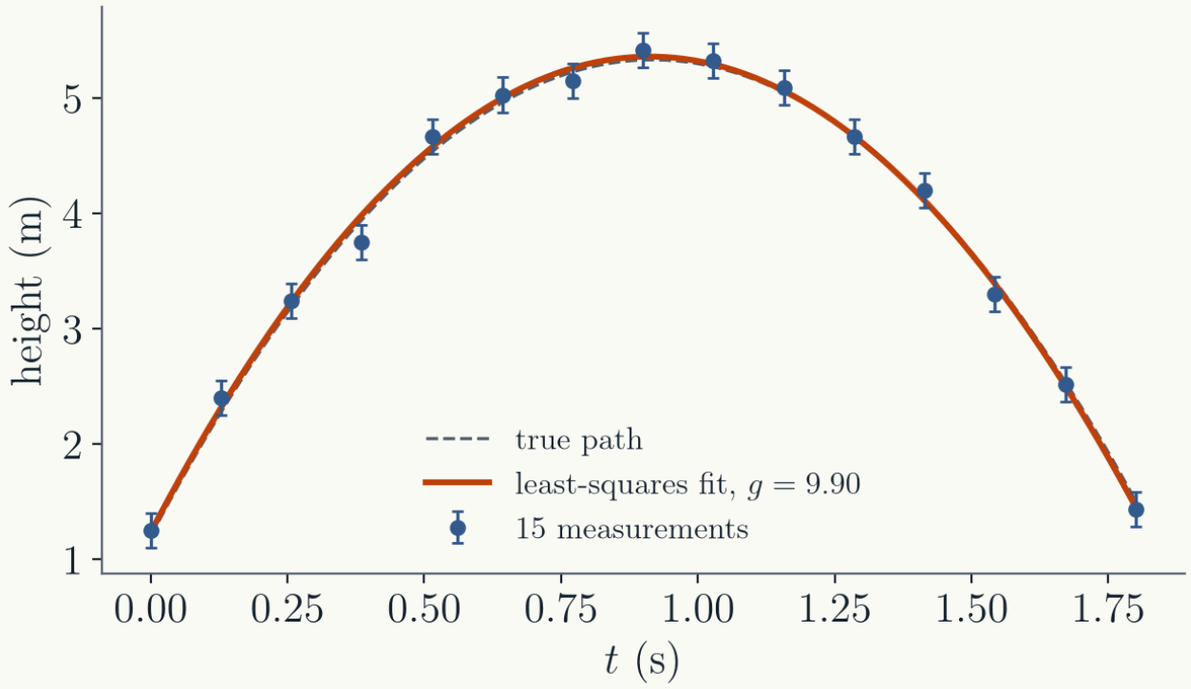
Least squares fitting

15 measurements of a thrown ball's height. The model is a parabola with three unknown coefficients:

$$y(t) = a_0 + a_1 t + a_2 t^2$$

One equation per measurement, $a_0 + a_1 t_i + a_2 t_i^2 = y_i$. That is $\mathbf{A}\mathbf{a} = \mathbf{y}$ with $\mathbf{A}$ 15×3: the rows are $(1,\ t_i,\ t_i^2)$.

Fifteen equations, three unknowns, noisy data: no exact solution. So take the closest one, $\mathbf{a} = \mathbf{A}^{+}\mathbf{y}$. It minimizes $|\mathbf{A}\mathbf{a} - \mathbf{y}|^2 = \sum_i \big(y(t_i) - y_i\big)^2$.



The fit, by the numbers

the matrix	$\mathbf{A}$ is 15×3, rows $(1,\ t_i,\ t_i^2)$; singular values $7.81,\ 2.33,\ 0.486;\ \kappa = 16$
rotate	$\mathbf{c} = \mathbf{U}^T\mathbf{y} = (-12.26,\ 8.70,\ -4.64)$: the data expressed in the three output directions
undo the stretch	$y_i = c_i/\sigma_i = (-1.57,\ 3.73,\ -9.56)$
rotate back	$\mathbf{a} = \mathbf{V}\mathbf{y} = (1.235,\ 9.034,\ -4.951)$, so $g = -2a_2 = 9.90$ (true 9.81, $v_0 = 9.03$, $y_0 = 1.24$)
check	$\lvert\mathbf{A}\mathbf{a} - \mathbf{y}\rvert = 0.33$, and $\mathbf{A}^T(\mathbf{A}\mathbf{a} - \mathbf{y}) = \mathbf{0}$ to rounding

The last line is the geometry of least squares: the leftover $\mathbf{A}\mathbf{a} - \mathbf{y}$ is perpendicular to every column of $\mathbf{A}$. Nothing the model can do would reduce it further.

Why not the normal equations?

$\mathbf{A}^T(\mathbf{A}\mathbf{a} - \mathbf{y}) = \mathbf{0}$ is a square system, $(\mathbf{A}^T\mathbf{A})\,\mathbf{a} = \mathbf{A}^T\mathbf{y}$, and LU solves it. The catch: the singular values of $\mathbf{A}^T\mathbf{A}$ are σ_i^2, so its condition number is κ^2.

polynomial degree	$\kappa(\mathbf{A})$	κ^2	residual, SVD	residual, normal equations
4	7×10^{2}	4×10^{5}	1.8×10^{-2}	1.8×10^{-2}
8	6×10^{5}	4×10^{11}	2.1×10^{-6}	2.1×10^{-6}
12	7×10^{8}	5×10^{17}	3.9×10^{-11}	4.7×10^{-8}
16	8×10^{11}	6×10^{23}	9.8×10^{-15}	2.7×10^{-6}

Fitting a smooth function on $[0, 1]$ with 60 points. Once κ^2 passes 10^{16}, the normal equations return coefficients that are wrong by a factor of order one; the SVD keeps improving.

Same lesson as Gram-Schmidt against Householder: work on $\mathbf{A}$ itself, never on a product that squares its conditioning.

In practice

Python: `lstsq` is the SVD route, and `rcond` is the cutoff for small σ_i:

```
A = np.vstack([np.ones_like(t), t, t**2]).T
a, res, rank, sv = np.linalg.lstsq(A, y, rcond=None)

U, s, Vt = np.linalg.svd(A, full_matrices=False)
```

C++ with Eigen: build the thin SVD and ask it to solve.

```
Eigen::MatrixXd A(15, 3);
Eigen::VectorXd y(15);       // fill both
auto svd = A.bdcSvd(Eigen::ComputeThinU
                  | Eigen::ComputeThinV);
Eigen::VectorXd a = svd.solve(y);
```

- For a well-conditioned $\mathbf{A}$, QR (`colPivHouseholderQr().solve`) does the same job faster. The SVD is for when you want to see the σ_i.
- Parameters that enter non-linearly ($e^{-t/\tau}$, $\sin \omega t$) need an iterative fit (`scipy.optimize.curve_fit`), which solves a linear least-squares problem like this one at every step.

Part III · Keeping only what matters

Low-rank approximation

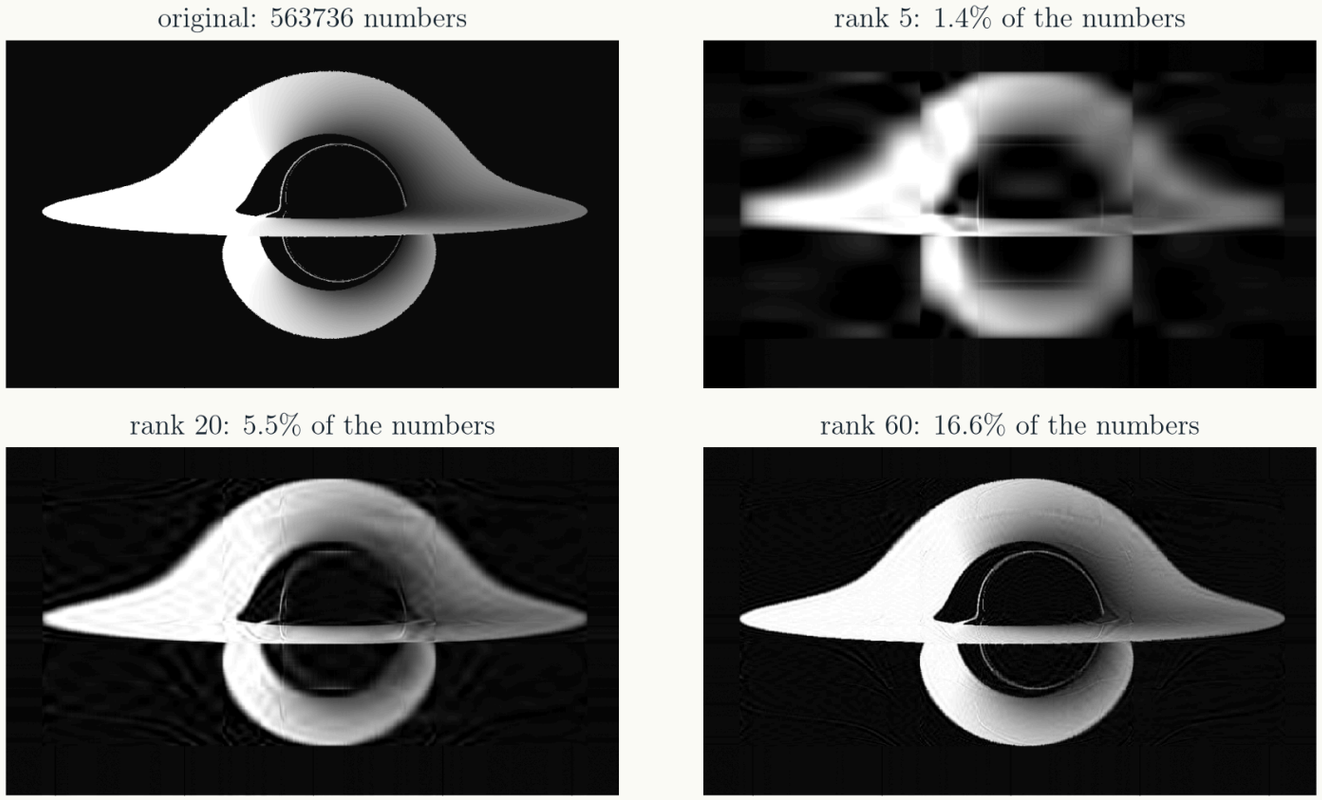
A matrix as a sum of layers

Written out column by column, the SVD is a sum of rank-one pieces, one per singular value:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \, \mathbf{u}_i \mathbf{v}_i^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots$$

- The σ_i are sorted, so the terms come in order of importance. Keep the first k: $\mathbf{A}_k = \sum_{i \le k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$.
- $\mathbf{A}_k$ is the *closest* matrix of rank k to $\mathbf{A}$, and the error is the first thing dropped: $\max_{|\mathbf{x}|=1} |(\mathbf{A} - \mathbf{A}_k)\mathbf{x}| = \sigma_{k+1}$.
- Storage: k vectors of length m, k of length n, and k numbers. $k(m + n + 1)$ instead of mn.

The Project 1 image at low rank



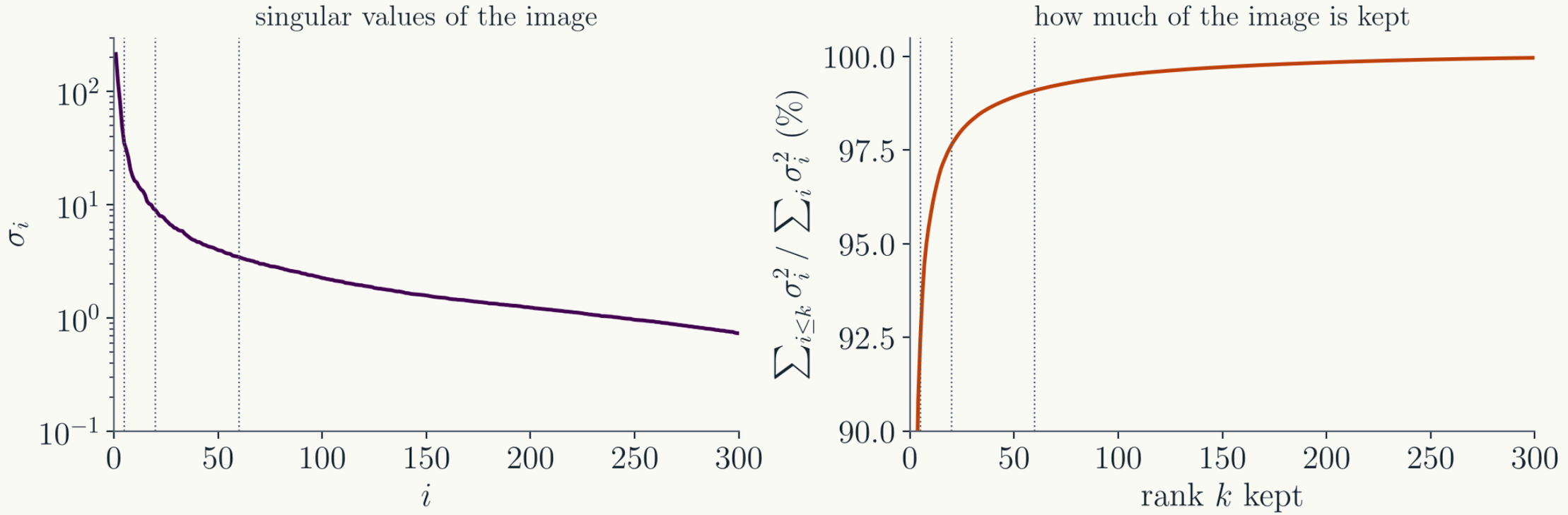
One color channel of the Project 1 image: 566×996 pixels, $563{,}736$ numbers. Treat it as a matrix and keep the first k layers.

Rank 5 stores 1.4% of the numbers and is a smear.

Rank 20 stores 5.5% and shows the shadow and the lensed arcs.

Rank 60 stores 17% and is hard to tell from the original.

Where the information is



- $\sigma_1 = 215$; the error at rank 5 is $\sigma_6 = 31$, at rank 20 it is $\sigma_{21} = 8.5$, at rank 60 it is 3.4. Rank 20 keeps 97.6% of $\sum \sigma_i^2$.
- A picture with structure has a few large σ_i and a long tail. Pure noise has a flat spectrum. So truncation keeps the structure and drops most of the noise.

Beyond pictures

Anything that is a matrix can be truncated the same way, and the σ_i tell you how much you lose:

- **Simulation snapshots** as columns: the first few $\mathbf{u}_i$ are the dominant modes of the flow or field, and a model built on them runs in a fraction of the time (reduced-order modelling).
- **A measured response matrix** (detector, instrument, linear inverse problem): small σ_i are the directions the measurement barely sees. Cutting them is regularization.
- **A two-particle wavefunction** ψ_{ij}, particle 1 in state i and particle 2 in state j: its SVD is the Schmidt decomposition, and the number of nonzero σ_i measures the entanglement.

Part IV · Finding the directions in data

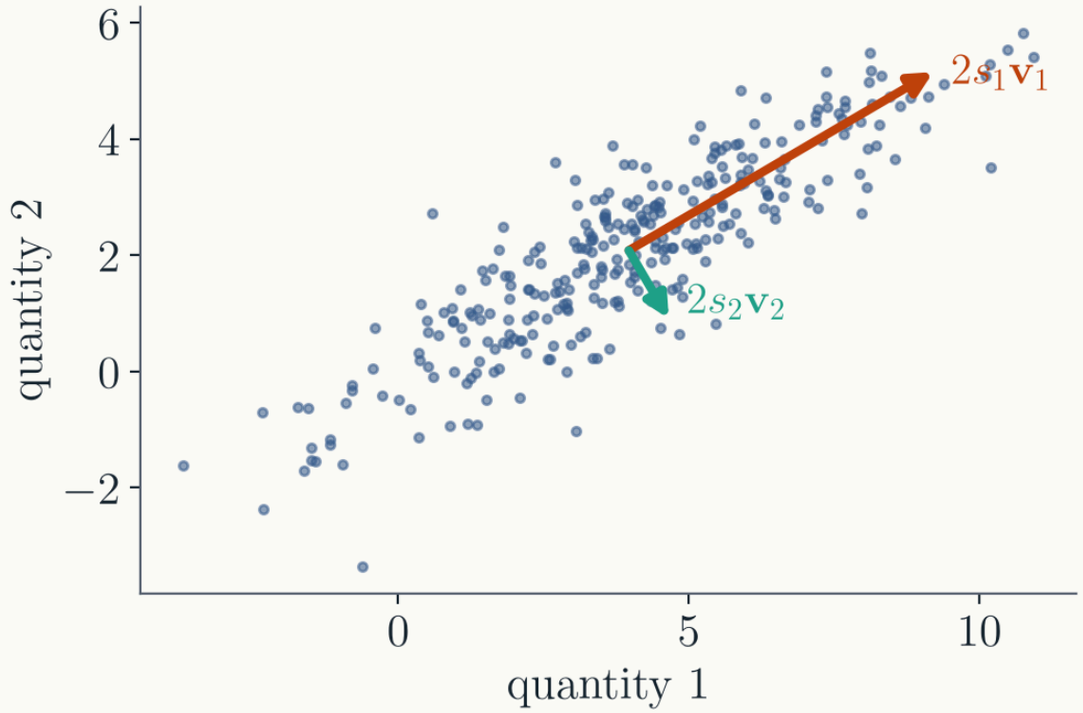
Principal component analysis

The data matrix

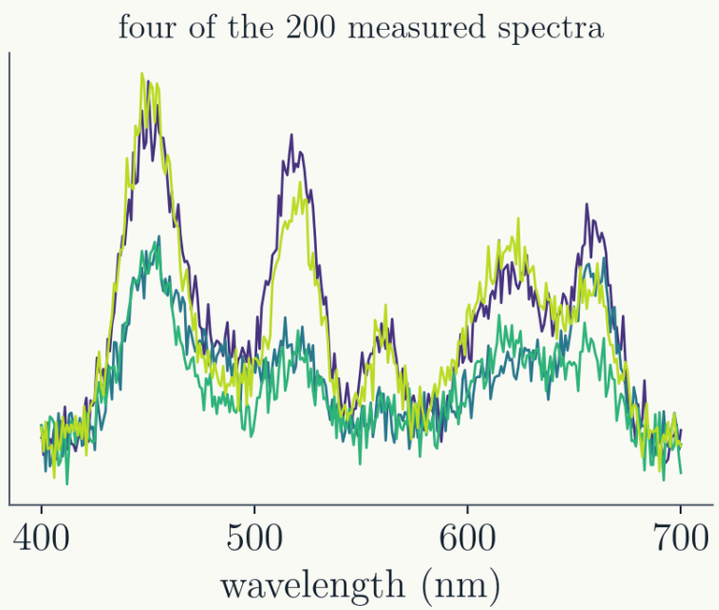
n measurements of p quantities: an $n \times p$ matrix $\mathbf{X}$, one row per measurement. Subtract the mean of each column.

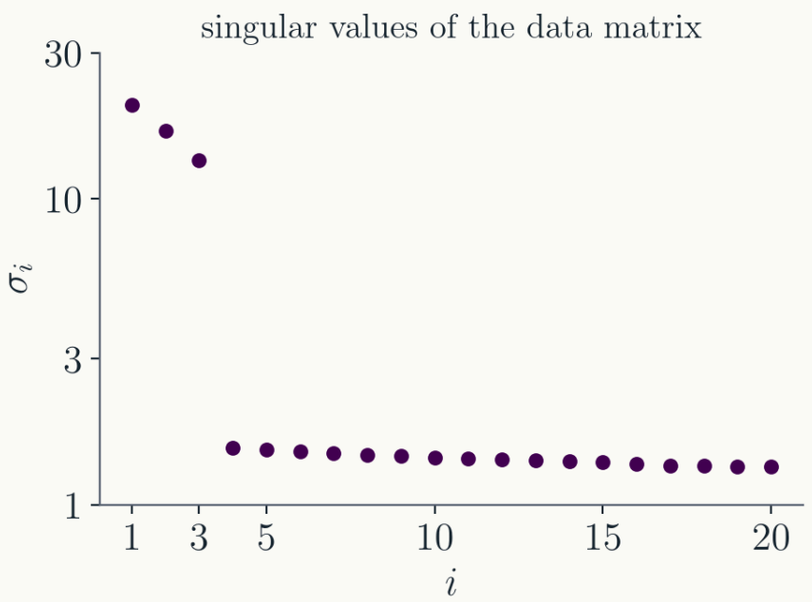
SVD: $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. The $\mathbf{v}_i$ are the *principal directions*, and the spread of the data along $\mathbf{v}_i$ is $s_i = \sigma_i / \sqrt{n-1}$.

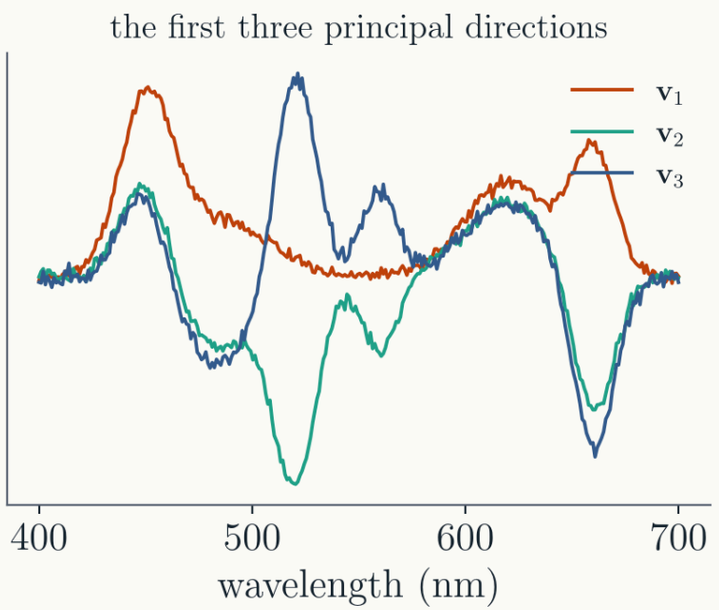
Here: 300 points. Along $\mathbf{v}_1$ the spread is 3.04, along $\mathbf{v}_2$ it is 0.68; 95% of the variance is along $\mathbf{v}_1$. (Generated with 3.0 and 0.7 at 30°; the SVD found 30.4°.)



Example: 200 spectra, 300 wavelengths







- Each spectrum is a mix of three emitting species in different amounts, plus noise. Nobody told the SVD that.
- Three singular values stand out ($20.2, 16.7, 13.3$), then a flat noise floor near 1.5. The data live in a 3-dimensional subspace of the 300-dimensional wavelength space, and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span it.

PCA and the covariance matrix

The covariance of the p quantities is the $p \times p$ matrix $\mathbf{C} = \mathbf{X}^T\mathbf{X}/(n-1)$. With the SVD of $\mathbf{X}$:

$$\mathbf{C} = \mathbf{V}\,\frac{\mathbf{\Sigma}^T\mathbf{\Sigma}}{n-1}\,\mathbf{V}^T$$

- So the principal directions are the eigenvectors of the covariance matrix, and the variances along them, s_i^2, are its eigenvalues. That is how PCA is usually introduced.
- Lecture 14 went the other way: from a covariance matrix, Cholesky built correlated random numbers. PCA reads the covariance's axes back out of the data.
- Computing $\mathbf{C}$ first squares the conditioning, exactly as the normal equations did. Take the SVD of $\mathbf{X}$ directly.

How the SVD is computed

- Not from $\mathbf{A}^T\mathbf{A}$: that squares κ and loses the small σ_i, which are often the point.
- Householder reflections from both sides bring $\mathbf{A}$ to *bidiagonal* form (two diagonals), the way they brought a symmetric matrix to tridiagonal form for eigenvalues.
- Then an iteration in the family of the QR iteration drives the off-diagonal to zero. The diagonal that remains holds the σ_i; the accumulated reflections and rotations are $\mathbf{U}$ and $\mathbf{V}$.
- Cost: a small multiple of QR's, about $4mn^2$ for a tall matrix. `numpy.linalg.svd` calls LAPACK's `gesdd`; Eigen has `BDCSVD` (large) and `JacobiSVD` (small, most accurate).

Golub and Kahan, 1965. You will never write it; you should know it is a factorization of $\mathbf{A}$ itself, with orthogonal transformations only, which is why it is accurate.

Summary

- $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ exists for every matrix: rotate, stretch, rotate. It reveals rank, range, nullspace, and the condition number σ_1/σ_n.
- **Closest solution:** $\mathbf{x} = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^T\mathbf{b}$ minimizes $|\mathbf{A}\mathbf{x} - \mathbf{b}|$. Least-squares fitting is this with $A_{ij} = f_j(x_i)$. Never form $\mathbf{A}^T\mathbf{A}$.
- **Low rank:** keep the first k terms of $\sum \sigma_i \mathbf{u}_i \mathbf{v}_i^T$; the error is σ_{k+1}.
- **PCA:** the SVD of the centered data matrix. The $\mathbf{v}_i$ are the directions of largest spread, $\sigma_i^2/(n-1)$ the variances.

Next time: partial differential equations. The advection equation, and why the obvious scheme blows up.